

Chitosan-Based Microspheres of Glibenclamide Incorporated in In-Situ Gel for Nasal Delivery: Mathematical Optimization, Kinetic Modeling, Ex-Vivo Permeation, and In-Vivo Bioavailability Assessment

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ABSTRACT

Background: Glibenclamide (GLB) is a second-generation anti-diabetic sulfonylurea with poor water solubility (18 µg/mL) and high first-pass hepatic metabolism (>50%), resulting in erratic oral bioavailability (~45%).

Objective: The objective of this investigation was to formulate, mathematically optimize, and evaluate a mucoadhesive, thermosensitive dual delivery system—glibenclamide-loaded chitosan microspheres embedded in a Pluronic-based in-situ gel—for intranasal administration.

Methods: Chitosan microspheres were synthesized via emulsion cross-linking using glutaraldehyde. A 3² Full Factorial Design was used to optimize chitosan concentration (X1) and glutaraldehyde volume (X2) against particle size (Y1), entrapment efficiency (Y2), and 8-hour drug release (Y3). The optimized microspheres (Fopt) were incorporated into an in-situ gel matrix of Pluronic F-127 (18% w/v) and PEG 400 (1.5% w/v). Kinetic drug release modeling, rheological profile, ex-vivo mucosal permeation, histopathology, and in-vivo PK/PD studies on streptozotocin-induced diabetic Wistar rats were conducted.

Results: Response surface methodology yielded Fopt (X1 = 1.5% w/v, X2 = 2.0 mL) with a mean particle diameter of 42.35 ± 2.15 µm, entrapment efficiency of 84.62 ± 1.85%, and 8-hour drug release of 81.15 ± 1.40%. R² values for regression models exceeded 0.98. The gel showed a phase transition temperature (T_{sol→gel}) of 33.4 ± 0.3 °C and mucoadhesive force of 4250 ± 112 dynes/cm². Release kinetics followed the Korsmeyer-Peppas model (n = 0.62), indicating non-Fickian anomalous diffusion. In-vivo PK revealed C_{max} = 142.50 ± 8.20 ng/mL and AUC_{0-inf} = 1485.20 ± 62.40 ng*h/mL for nasal gel versus C_{max} = 58.20 ± 4.10 ng/mL and AUC_{0-inf} = 412.30 ± 28.10 ng*h/mL for oral suspension.

Conclusion: The nasal microsphere in-situ gel exhibited a 3.6-fold increase in relative bioavailability (F_{rel} = 360.22%), successfully bypassing hepatic pre-systemic metabolism and providing sustained blood glucose control over 24 hours.

Keywords: Glibenclamide, Chitosan Microspheres, Design of Experiments (DoE), 3² Factorial Design, Korsmeyer-Peppas Kinetics, Intranasal Bioavailability, In-Situ Gel.

1. INTRODUCTION

Type-2 Diabetes Mellitus (T2DM) is a chronic metabolic disorder requiring long-term glycemic control. Glibenclamide (GLB), a lipophilic BCS Class II drug ($\log P = 3.75$), binds to ATP-sensitive potassium (K_{ATP}) channel receptors on pancreatic beta-cells to stimulate insulin release.

Despite high potency, oral GLB therapy faces significant bio-pharmaceutical limitations:

1. Low Aqueous Dissolution Rate: Aqueous solubility is extremely low (0.018 mg/mL at pH 7.4), making dissolution the rate-limiting step for absorption.
2. First-Pass Hepatic Extraction: Extensive metabolism via hepatic cytochrome P450 isoenzymes (CYP2C9) limits systemic availability to 45-50%.
3. Fluctuating Plasma Concentration: Transient blood peaks cause sudden hypoglycemic episodes, followed by sub-therapeutic troughs.

The nasal cavity provides a non-invasive administration site with a highly vascularized mucosal surface ($\sim 150 \text{ cm}^2$), avoiding hepatic pre-systemic metabolism and providing direct entry to systemic circulation. However, mucociliary clearance (MCC) clears simple liquid solutions within 15-20 minutes.

To overcome MCC, this study developed a dual-retention system:

- Chitosan Microspheres: Polymer cross-linking traps the drug while cationic primary amines ($-\text{NH}_3^+$) bind electrostatically to anionic mucosal sialic acid residues, opening tight junctions (ZO-1) reversibly.
- Thermo-Responsive In-Situ Gel Base: Pluronic F-127 solution remains liquid at room temperature (22-25 °C) for easy application, converting to a semi-solid matrix at physiological nasal temperature (32-34 °C).

2. MATERIALS AND METHODS

2.1 Reagents and Materials

Glibenclamide (USP, >99.2% purity) was obtained from Sun Pharmaceutical Industries Ltd. Low Molecular Weight Chitosan (Degree of Deacetylation >85%, molecular weight 150 kDa), Pluronic F-127 (Poloxamer 407), Polyethylene Glycol 400 (PEG 400), and Glutaraldehyde (25% v/v) were sourced from Sigma-Aldrich. Light liquid paraffin, Span 80, and glacial acetic acid were purchased from Merck India Ltd.

2.2 Synthesis of Glibenclamide-Loaded Chitosan Microspheres

Microspheres were produced using a w/o emulsion cross-linking protocol:

1. Aqueous Phase: 1.0-2.0% w/v Chitosan was dissolved in 1.5% v/v aqueous acetic acid. Glibenclamide (100 mg) dissolved in 2 mL methanol was mixed under probe sonication (2 minutes, 40 kHz).

2. Emulsification: The aqueous mixture was added dropwise to 50 mL light liquid paraffin containing 1.0% w/v Span 80, stirred at 1800 RPM for 30 minutes to form a stable w/o emulsion.
3. Chemical Cross-Linking: Glutaraldehyde solution (1.0-3.0 mL) was introduced dropwise. Cross-linking was maintained at 40 °C for 3.5 hours.
4. Purification: Microspheres were isolated by centrifugation (4500 RPM, 15 min), washed three times with petroleum ether to remove paraffin, washed with deionized water, and vacuum-dried at 40 °C for 24 hours.

3. MATHEMATICAL OPTIMIZATION & STATISTICAL CALCULATIONS (DoE)

3.1 Design of Experiments (DoE) Equations

A 3² Full Factorial Design evaluated two formulation variables at three levels.

Factor Coding & Real Values:

- X1 (Chitosan Conc. % w/v): Low (-1) = 1.0%, Medium (0) = 1.5%, High (+1) = 2.0%
- X2 (Glutaraldehyde Vol. mL): Low (-1) = 1.0 mL, Medium (0) = 2.0 mL, High (+1) = 3.0 mL

Second-Order Polynomial Response Equation:

$$Y = B_0 + B_1 * X_1 + B_2 * X_2 + B_{12} * X_1 * X_2 + B_{11} * X_1^2 + B_{22} * X_2^2$$

3.2 Matrix of Experimental Factorial Runs & Observed Data

(*F5 = Optimized Formulation Fopt)

3.3 Regression Equations:

- Particle Size (Y1) = 42.61 + 8.92*X1 + 3.81*X2 + 1.25*X1*X2 + 1.15*X1² - 0.42*X2²
- Entrapment Efficiency (Y2) = 83.94 + 10.15*X1 + 4.82*X2 - 0.95*X1*X2 - 3.12*X1² - 1.85*X2²
- Q8 Drug Release (Y3) = 80.22 - 11.45*X1 - 8.92*X2 - 1.10*X1*X2 - 2.85*X1² + 0.15*X2²

3.4 ANOVA Evaluation for Response Surface Models (Entrapment Efficiency Y2)

(R² = 0.9930, Adjusted R² = 0.9813, Adequate Precision = 28.42)

Table 1. Matrix of experimental factorial runs and observed data.

F1	(-1, -1)	1.0	1.0	32.14 ± 1.20	62.30 ± 1.40	92.40 ± 2.10
F2	(-1, 0)	1.0	2.0	35.80 ± 1.50	68.50 ± 1.10	85.10 ± 1.90
F3	(-1, +1)	1.0	3.0	38.20 ± 1.10	71.20 ± 1.80	74.30 ± 2.00
F4	(0, -1)	1.5	1.0	39.40 ± 1.40	74.10 ± 1.30	88.20 ± 1.50
F5*	(0, 0)	1.5	2.0	42.35 ± 1.20	84.62 ± 1.85	81.15 ± 1.40
F6	(0, +1)	1.5	3.0	46.10 ± 1.80	86.40 ± 1.60	68.90 ± 1.80
F7	(+1, -1)	2.0	1.0	48.50 ± 2.00	81.20 ± 1.70	79.40 ± 2.20
F8	(+1, 0)	2.0	2.0	52.10 ± 1.60	88.30 ± 1.50	64.20 ± 1.60
F9	(+1, +1)	2.0	3.0	58.90 ± 2.10	91.10 ± 1.90	52.80 ± 1.30

3.3 Regression Equations

Particle Size (Y_1) = $42.61 + 8.92 \cdot X_1 + 3.81 \cdot X_2 + 1.25 \cdot X_1 \cdot X_2 + 1.15 \cdot X_1^2 - 0.42 \cdot X_2^2$

Entrapment Efficiency (Y_2) = $83.94 + 10.15 \cdot X_1 + 4.82 \cdot X_2 - 0.95 \cdot X_1 \cdot X_2 - 3.12 \cdot X_1^2 - 1.85 \cdot X_2^2$

Q8 Drug Release (Y_3) = $80.22 - 11.45 \cdot X_1 - 8.92 \cdot X_2 - 1.10 \cdot X_1 \cdot X_2 - 2.85 \cdot X_1^2 + 0.15 \cdot X_2^2$

Table 2. ANOVA evaluation for the response surface model of entrapment efficiency.

Model	742.18	5	148.44	84.58	< 0.0001 (Significant)
X1 (Chitosan)	618.15	1	618.15	352.22	< 0.0001
X2 (Glutaraldehyde)	139.39	1	139.39	79.42	0.0028
X1*X2	3.61	1	3.61	2.06	0.2461
X1 ²	19.47	1	19.47	11.10	0.0447
X2 ²	6.85	1	6.85	3.90	0.1432
Residual	5.26	3	1.75	--	--
Total	747.44	8	--	--	--

($R^2 = 0.9930$, Adjusted $R^2 = 0.9813$, Adequate Precision = 28.42)

4. CHARACTERIZATION OF OPTIMIZED FORMULATION (Fopt)

4.1 Particle Size and Zeta Potential

- Mean Particle Diameter: $42.35 \pm 2.15 \mu\text{m}$
- Polydispersity Index (PDI): 0.214
- Zeta Potential: $+28.4 \pm 1.8 \text{ mV}$

4.2 In-Vitro Dissolution Kinetics & Mathematical Release Modeling

4.0 54.80 ± 1.80%

- Zero-Order: $R^2 = 0.9215$, $K = 7.42 \text{ \%}/\text{h}$
- First-Order: $R^2 = 0.9652$, $K = 0.211 \text{ h}^{-1}$
- Higuchi Model: $R^2 = 0.9812$, $K = 26.85 \text{ \%}/\text{h}^{0.5}$
- Korsmeyer-Peppas: $R^2 = 0.9945$, $K = 23.14 \text{ \%}/\text{h}^n$, $n = 0.62$ (Non-Fickian Anomalous Diffusion)
- Hixson-Crowell: $R^2 = 0.9482$, $K = 0.082 \text{ mg}^{1/3}/\text{h}$

Table 3. In-vitro cumulative drug release profile of the optimized formulation.

0.5	$12.45 \pm 0.85\%$
1.0	$22.10 \pm 1.15\%$
2.0	$35.40 \pm 1.40\%$
4.0	$54.80 \pm 1.80\%$
6.0	$69.20 \pm 1.65\%$
8.0	$81.15 \pm 1.40\%$
10.0	$89.50 \pm 1.90\%$
12.0	$95.20 \pm 1.30\%$

4.3 Kinetic Model Fitting

- Zero-Order: $R^2 = 0.9215$, $K = 7.42\%/\text{h}$
- First-Order: $R^2 = 0.9652$, $K = 0.211\text{ h}^{-1}$
- Higuchi Model: $R^2 = 0.9812$, $K = 26.85\%/\text{h}^{0.5}$
- Korsmeyer-Peppas: $R^2 = 0.9945$, $K = 23.14\%/\text{h}^n$, $n = 0.62$ (Non-Fickian Anomalous Diffusion)
- Hixson-Crowell: $R^2 = 0.9482$, $K = 0.082\text{ mg}^{1/3}/\text{h}$

5. IN-SITU GEL FORMULATION & CHARACTERIZATION

5.1 Gel Base Properties

- Sol-to-Gel Transition Temperature ($T_{\text{sol} \rightarrow \text{gel}}$): $33.4 \pm 0.3\text{ }^\circ\text{C}$
- Mucoadhesive Strength: $4250 \pm 112\text{ dynes/cm}^2$

5.2 Viscosity Measurements Across Shear Rates

5.3 Trans-Mucosal Permeation (Goat Nasal Mucosa)

Table 4. Viscosity measurements of the in-situ gel across temperature and shear rate.

Temperature (°C)	State	Shear Rate (1/s)	Viscosity (Pa·s)
25.0	Sol	10	145 ± 8
25.0	Sol	100	142 ± 6
34.0	Gel	10	4850 ± 120
34.0	Gel	50	1920 ± 85
34.0	Gel	100	1110 ± 45

Table 5. Ex-vivo trans-mucosal permeation parameters.

Parameter	Formulation A	Formulation B	Ratio (B/A)
Cumulative Drug Permeated at 12h	$24.12 \pm 1.85\%$	$82.45 \pm 2.10\%$	3.42-fold
Steady-State Flux (Jss)	$10.05\text{ }\mu\text{g/cm}^2\cdot\text{h}$	$34.35\text{ }\mu\text{g/cm}^2\cdot\text{h}$	341-fold
Permeability (Papp)	2.01×10^{-3}	6.87×10^{-3}	3.41-fold
Lag Time (t _{lag})	1.42 h	0.25 h	5.68-fold faster

6. IN-VIVO PHARMACOKINETIC (PK) & PHARMACODYNAMIC (PD) EVALUATION

6.1 Pharmacokinetic Parameters Summary

6.2 Pharmacodynamic Outcome

- Oral Suspension: Peak blood glucose reduction of 38.5% at 2.5 h; returned to baseline within 10 h.
- Intranasal Gel Matrix: Peak glucose reduction of 58.4% at 4 h; maintained >40% glucose reduction up to 18 h post-administration.

Table 6. Pharmacokinetic parameters following oral and intranasal administration.

C _{max} (ng/mL)	58.20 ± 4.10	142.50 ± 8.20	p < 0.001
T _{max} (h)	2.00 ± 0.25	1.00 ± 0.10	p < 0.01
k _{el} (h ⁻¹)	0.220 ± 0.02	0.101 ± 0.01	p < 0.01
Elimination t _{1/2} (h)	3.15 ± 0.30	6.85 ± 0.45	p < 0.01
AUC _{0-24h} (ng*h/mL)	395.10 ± 22.10	1380.40 ± 54.30	p < 0.001
AUC _{0-inf} (ng*h/mL)	412.30 ± 28.10	1485.20 ± 62.40	p < 0.001
Relative Bioavailability	100.00%	360.22%	p < 0.001

7. HISTOPATHOLOGICAL SAFETY EVALUATION

- Control Mucosa (PBS pH 6.4): Intact columnar epithelium with normal ciliated border.
- Formulated Gel Treated Mucosa: Preserved ciliated border, absence of mucosal erosion, sub-mucosal hemorrhage, necrosis, or inflammatory infiltration.

8. CONCLUSION

The chitosan microsphere-loaded in-situ gel formulation was successfully optimized and evaluated. The dual delivery system prolonged nasal residence, enhanced trans-mucosal absorption, provided non-Fickian drug release, increased oral bioavailability by 3.6-fold, and demonstrated high mucosal safety in animal models.

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